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SIGNIFICANCE OF ANOMALOUS THERMOREGULATION
IN THE PRE-DIABETIC SPINY MOUSE (*ACOMYS CAHIRINUS*):
OXYGEN CONSUMPTION AND TEMPERATURE REGULATION

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Summary. In keeping with studies of other desert rodents, the potentially diabetic spiny mouse has been demonstrated to have a very low basal metabolic rate, disproportionate to its body weight. The maintenance of a lower body temperature in response to high environmental temperatures and a lack of increase in metabolic rate in response to cooling have also been demonstrated. Assessments of "mechanical efficiency" have shown that spiny mice carry a potential selective advantage under fasting conditions. The findings, some of which are similar to those noted in other desert rodents, and in other species showing either spontaneous or induced hyperglycaemia, suggest that the low metabolic rate is at least partly based on ineffective glucose utilization. This phenomenon may be the common denominator of the survival advantage which has allowed both the successful evolution of species inhabiting warm, arid climates, and the persistence of the diabetic genotype in animal and human populations.

INTRODUCTION.

Substantial literature has accumulated in the last 20 years on the thermoregulation of rodents from desert and other environments. Murie (1961) and McNab and Morrison (1963) have suggested that the low oxygen consumption and proportionately lower body temperature now demonstrated for a variety of desert rodents represent a mechanism of selective value for avoidance of lethal body temperatures, for reducing the amount of water required for evaporative cooling and for minimizing caloric requirement. The mechanism by which these characteristics have evolved has received comparatively little attention.

A number of rodents, including *Psammomys obesus* (Schmidt-Nielsen, Haines and Hackel, 1964), *Acomys cahirinus* (Gonet, Mouglin and Renold, 1965), *Ctenomys talarum* (Wise *et al.*, 1972) and *Notomys fuscus* (Wise and Watts, unpublished observations) have a natural habitat in warm, comparatively arid

climates, and have been shown to have a high prevalence of diabetes mellitus when brought into captivity. The purpose of the following studies was to characterize the thermoregulatory responses in *Acomys cahirinus* (spiny mouse), with a view to examining possible relationships to its potential diabetic status. It was hoped that such an evaluation might contribute to understanding both of the apparently anomalous thermoregulation seen in some species and of the evolutionary significance of the diabetic genotype.

The spiny mouse is known to develop obesity and often fatal diabetes in high prevalence (20%) under conditions of free feeding (Gonet *et al.*, 1965; Hefti and Flückiger, 1967), while food restriction will maintain normoglycaemia. Gonet *et al.* (1966) have demonstrated that pancreatic islet hyperplasia is a universal finding in all spiny mice studied so that potential diabetic status appears to be a species-wide characteristic. The spiny mouse is widely distributed in arid areas of Africa, Egypt, Crete, Cyprus, Morocco and Israel. A laboratory colony was established by Flückiger and Operschall (1962) from original stock inhabiting arid zones of Israel, and used for studies of neurohypophyseal morphology. The animals used for the present studies were obtained from the parent colony of Flückiger, some 14 generations removed from the original breeding stock.

MATERIALS AND METHODS.

Animals.

Five breeding triplets (including 2 females, 1 male) were housed in separate plastic boxes measuring 35 x 25 x 15 cm and bedded on straw. Subsequent litters were weaned at 3-4 weeks and housed individually or formed into new breeding triplets. To avoid any anomaly due to overt diabetes itself, all animals were maintained on a diet consisting of carrot and lettuce, sunflower seeds and a relatively inaccessible supply of synthetic diet pellets (Dixon 41/B) in an overhead trough, since Hefti and Flückiger (1967) have shown that ease of access to food is critical to the development of both obesity and hyperglycaemia. In practice, on the above régime, animals became neither obese nor overtly diabetic. That they were in fact potentially diabetic was revealed by mild hyperglycaemia when ready access was allowed to diet pellets, and by subsequent pancreatic histology, which demonstrated universal and typical islet hypertrophy and hyperplasia.

Control animals were standard laboratory bred white (Swiss strain) mice, housed under identical conditions but with a food source only of mouse pellets (Dixon 41/B). All animals were kept in a room with an ambient temperature of $21 \pm 1^\circ$. Spiny mice used for the following studies were males weighing 32 to 41 g at age 5-7 months: white control mice were also males, and weighed 26-44 g within the same age range.

Oxygen consumption measurement.

A closed circuit method was used, adapted from MacLagan and Sheahan (1949). The mouse to be studied was placed on a perforated zinc grid in a 1300 ml glass dessicating chamber, the floor of which was covered to a depth of 1 cm with anaesthetic quality soda-lime (Calona). A small receptacle on the floor of the dessicator contained anhydrous calcium chloride, and both reagents were renewed after each experiment. A thermometer suspended from the roof of the chamber recorded ambient temperature, and the entire chamber was submerged in water within a glass tank the circulation and temperature of which was controlled by a thermostatically regulated heater and agitator capable of rapid, accurate adjustment within the desired range of 23° to 39° . The dessicator outlet was connected via a

3-way junction to a vertical tube manometer and a syringe for calibration. Twenty minutes' equilibration was allowed at each temperature. Manometer readings were recorded at 1 min intervals beginning after equilibration had occurred and continuing for approx. 20 min. Murie (1961) has shown that such a period is adequate for equilibration in terms of oxygen consumption. Sampling of chamber air content at the conclusion of each run revealed CO₂ concentrations never exceeding 0.2 vol% (Lloyd, 1958) in keeping with findings of MacLagan and Sheahan (1949) and Murie (1961) and indicative of effective CO₂ absorption.

Manometer readings from each run were plotted against time and the line of best fit drawn for the final 10 min of readings, from the slope of which pressure change per unit time was derived. Oxygen consumption was calculated from the following:—

$$O_{GH} = \frac{V}{W} \times \frac{p}{13.6 \times 760} \times \frac{t}{(273 + t)} \times \frac{1}{T}$$

where V is the chamber volume (ml), p = pressure change (mm H₂O/h), t = chamber temperature (°), T = observation time (h), and W = animal weight (g).

Activity of each animal was recorded each minute as resting (1 point), cleaning and other movements not associated with change of posture (2 points) and walking or other movements involving change of posture (3 points). A mean index of activity (A.I.) was derived for each complete temperature run including the equilibration period. Rectal temperature was recorded immediately on concluding each study using a thermistor probe inserted to a depth of 1 cm and recording on a direct reading potentiometer.

For each animal, studies were carried out within one day (0900 to 1600 h) from 23° to 39°. Each animal was studied both when fully fed (food withdrawn at 0800 h) and when fasting (food, but not water, withdrawn from 0800 h the previous day). To minimize artefactual psychological adaptation (Bramante, 1961), in half of the animals the studies in the fasted state preceded those in the fully-fed state. In each animal a period of one or two weeks separated the two studies. At the conclusion of these and further studies, some spiny and white mice were killed for assessment of body fat and for pancreatic histology.

Group statistics throughout are expressed as the mean $\pm$ standard error of the mean of 7.9 animals, with statistical analysis using Student's *t*-test.

RESULTS.

The cumulative data for oxygen consumption, activity index and rectal temperature are shown in Table 1, listed according to nutritional state and ambient temperature.

In fully-fed animals, white mice had a minimum oxygen consumption of 1.27 ± 0.08 ml O₂ g⁻¹ h⁻¹, and the zone of thermal neutrality extended from 29° to 35°. Spiny mice had corresponding values of 0.64 ± 0.03 ml O₂ g⁻¹ h⁻¹ ($t = 5.39$, $p < 0.001$) within a thermoneutral range of 33° to 37°. It has been shown by Brody (1945) that minimum oxygen uptake is inversely related to the 0.27 power of weight at the interspecific level ($M(\text{ml O}_2 \text{ g}^{-1} \text{ h}^{-1}) = 3.8 W^{-0.27}$, where W is body weight in grams). Since mean weights of fully-fed white and spiny mice were identical (35.6 ± 2.1 g, 35.9 ± 2.3 g), this calculated expected minimum (1.44 ml O₂ g⁻¹ h⁻¹) for both species is represented as one horizontal line in Fig. 1.

Decreasing ambient temperatures were associated with approx. parallel increments of oxygen consumption within the two groups. However, values of oxygen consumption in spiny mice were consistently lower at all temperatures, most marked above 35°. Rectal temperatures were measured only in fully-fed

animals (Table 1 and Fig. 2). At all ambient temperatures, mean rectal temperature (T_b) was lower in spiny mice, the difference being most marked at and above 33° , and significant ($p \leq 0.05$) at all ambient temperatures except 27° .

TABLE 1.
Oxygen consumption, rectal temperature, activity index (A.I.) and E-ratio in fed and fasted white and spiny mice.

	O ₂ Consumption (ml/g ⁻¹ /h ⁻¹)	Rectal T. ($^\circ$)	A.I. (Units)	E-Ratio (Units)
White mice fully-fed				
23 $^\circ$	2.28 $\pm$ 0.09	36.1 $\pm$ 0.2	2.5 $\pm$ 0.09	1.13 $\pm$ 0.06
25	1.86 $\pm$ 0.19	36.0 $\pm$ 0.2	2.0 $\pm$ 0.20	1.07 $\pm$ 0.09
27	1.71 $\pm$ 0.13	36.1 $\pm$ 0.2	1.7 $\pm$ 0.10	1.02 $\pm$ 0.04
29	1.36 $\pm$ 0.14	35.3 $\pm$ 0.2	1.4 $\pm$ 0.21	0.99 $\pm$ 0.03
31	1.27 $\pm$ 0.08	35.4 $\pm$ 0.2	1.4 $\pm$ 0.18	1.11 $\pm$ 0.07
33	1.42 $\pm$ 0.06	36.7 $\pm$ 0.3	1.9 $\pm$ 0.24	1.29 $\pm$ 0.15
35	1.41 $\pm$ 0.09	37.7 $\pm$ 0.4	2.1 $\pm$ 0.26	1.43 $\pm$ 0.14
37	1.54 $\pm$ 0.09	39.0 $\pm$ 0.1	1.9 $\pm$ 0.25	1.23 $\pm$ 0.09
39	1.56 $\pm$ 0.08	39.0 $\pm$ 0.1	1.4 $\pm$ 0.07	9.03 $\pm$ 0.05
White mice fasted				
23 $^\circ$	1.92 $\pm$ 0.06	—	2.7 $\pm$ 0.08	1.36 $\pm$ 0.07
27	1.25 $\pm$ 0.10	—	1.7 $\pm$ 0.15	1.33 $\pm$ 0.08
31	1.06 $\pm$ 0.09	—	1.4 $\pm$ 0.18	1.26 $\pm$ 0.08
35	1.33 $\pm$ 0.17	—	1.9 $\pm$ 0.33	1.41 $\pm$ 0.12
39	1.55 $\pm$ 0.10	—	2.1 $\pm$ 0.18	1.37 $\pm$ 0.11
Spiny mice fully-fed				
23 $^\circ$	1.64 $\pm$ 0.14	35.6 $\pm$ 0.2	2.6 $\pm$ 0.26	1.58 $\pm$ 0.09
25	1.38 $\pm$ 0.12	35.2 $\pm$ 0.4	1.9 $\pm$ 0.24	1.34 $\pm$ 0.11
27	1.11 $\pm$ 0.18	35.8 $\pm$ 0.4	1.8 $\pm$ 0.27	1.48 $\pm$ 0.23
29	0.92 $\pm$ 0.13	34.8 $\pm$ 0.2	1.4 $\pm$ 0.11	1.28 $\pm$ 0.10
31	0.95 $\pm$ 0.10	34.8 $\pm$ 0.2	1.2 $\pm$ 0.05	1.31 $\pm$ 0.14
33	0.75 $\pm$ 0.06	35.6 $\pm$ 0.3	1.2 $\pm$ 0.08	1.63 $\pm$ 0.09
35	0.64 $\pm$ 0.03	36.0 $\pm$ 0.2	1.1 $\pm$ 0.03	1.42 $\pm$ 0.14
37	0.75 $\pm$ 0.07	36.9 $\pm$ 0.4	1.3 $\pm$ 0.11	1.84 $\pm$ 0.11
39	1.11 $\pm$ 0.06	38.3 $\pm$ 0.2	2.0 $\pm$ 0.24	1.82 $\pm$ 0.17
Spiny mice fasted				
23 $^\circ$	0.87 $\pm$ 0.04	—	1.5 $\pm$ 0.19	1.69 $\pm$ 0.18
27	0.58 $\pm$ 0.03	—	1.1 $\pm$ 0.05	2.01 $\pm$ 0.14
31	0.53 $\pm$ 0.03	—	1.0 $\pm$ 0.01	1.93 $\pm$ 0.11
35	0.49 $\pm$ 0.02	—	1.1 $\pm$ 0.05	2.23 $\pm$ 0.12
39	0.94 $\pm$ 0.07	—	1.8 $\pm$ 0.17	2.09 $\pm$ 0.35

In fasted animals, a similar significant difference was noted. Mean minimum white mouse oxygen consumption was 1.06 ± 0.09 ml O₂ g⁻¹ h⁻¹ and for spiny mice 0.45 ± 0.02 ml O₂ g⁻¹ h⁻¹ ($t = 7.03$, $p < 0.001$). However, at lower ambient temperatures, oxygen consumption of fasted spiny mice remained low, a phenomenon not seen in fasted white mice. In response to fasting, white mice lowered oxygen consumption at thermoneutrality (31°) by a mean 16.5%; by comparison, reduction of spiny mice oxygen consumption at thermoneutrality (35°) was 23%. Within each species, activity indices at thermoneutrality were virtually identical in the two nutritional states.

Mean activity index trends at each ambient temperature were similar to those of oxygen consumption (Table 1, Fig. 3). However, compared with white mice, spiny mice demonstrated lower activity at thermoneutral temperatures: thus, the possibility existed that the apparently lower metabolic rates of spiny mice were dependent solely on a behavioural response involving relative immobility predominantly at higher temperatures. Two approaches were used to clarify this situation.

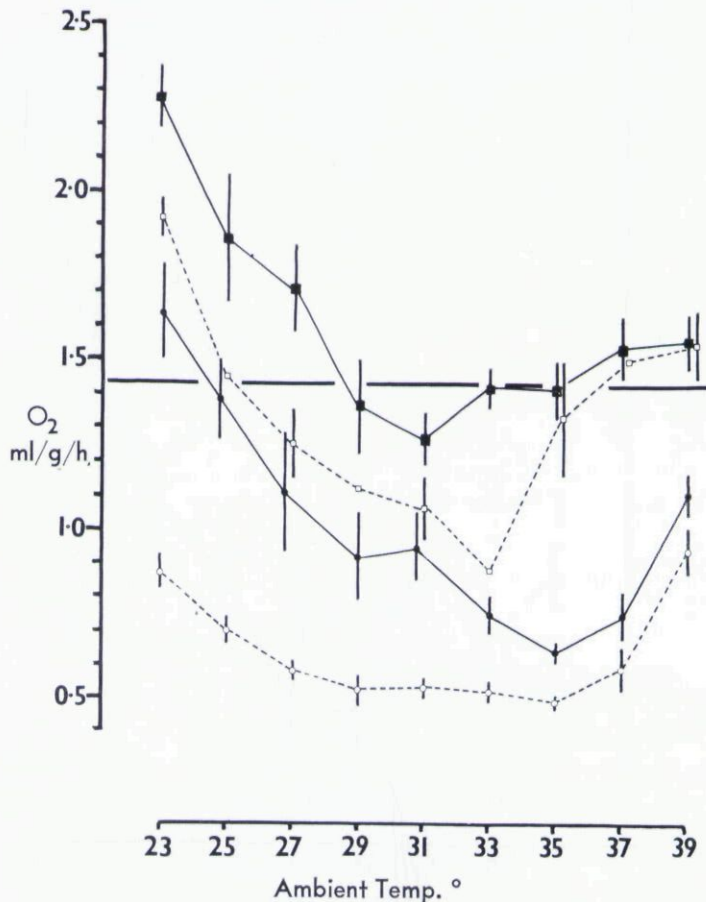


Fig. 1. Oxygen consumption (mean $\pm$ S.E.M.) at varying ambient temperatures in white and spiny mice: (horizontal line depicts expected minimum metabolic rate for both species in the fully-fed state).

■ white mice fully-fed ● spiny mice fully-fed
 □ white mice fasted ○ spiny mice fasted

First, mean values were derived for white mice using only runs with activity indices of 1.0, thereby ensuring a common denominator of physical immobility. On this basis, mean oxygen consumptions of 0.74 ± 0.10 and 1.06 ± 0.15 were obtained for spiny and white mice respectively ($p = 0.05$). Second,

an expression was derived relating spontaneous activity (A.I.) to oxygen consumption (O_{GH}). The ratio E was calculated for each animal where $E = \frac{A.I.}{O_{GH}}$ and, as noted by Bullard (1966), this ratio reflects the mechanical efficiency of muscular activity. Group means for each ambient temperature are listed in

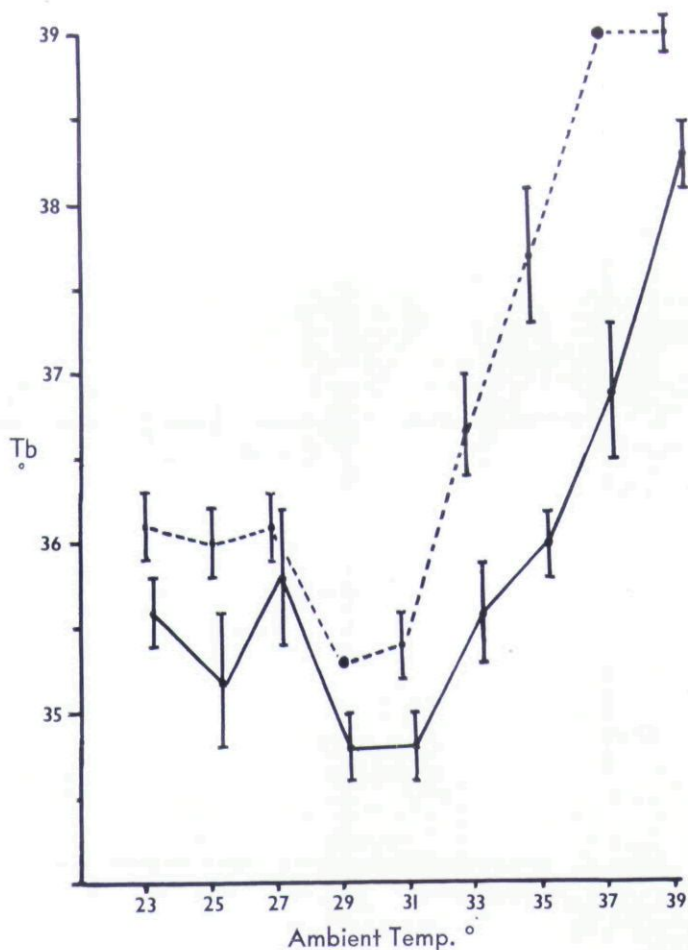


Fig. 2. Rectal temperature (mean $\pm$ S.E.M.) at different ambient temperatures in white and spiny mice.

---- white ——— spiny

Table 1. For fully-fed spiny mice, $E = 1.54 \pm 0.07$, significantly higher than for fully-fed white mice, 1.11 ± 0.02 ($t = 6.02$, $p < 0.001$). Corresponding values in fasted animals were 2.00 ± 0.08 and 1.3 ± 0.05 ($t = 6.18$, $p < 0.001$). Within each species, E -ratio was significantly higher in the fasted than in the fully-fed state ($p = 0.001$).

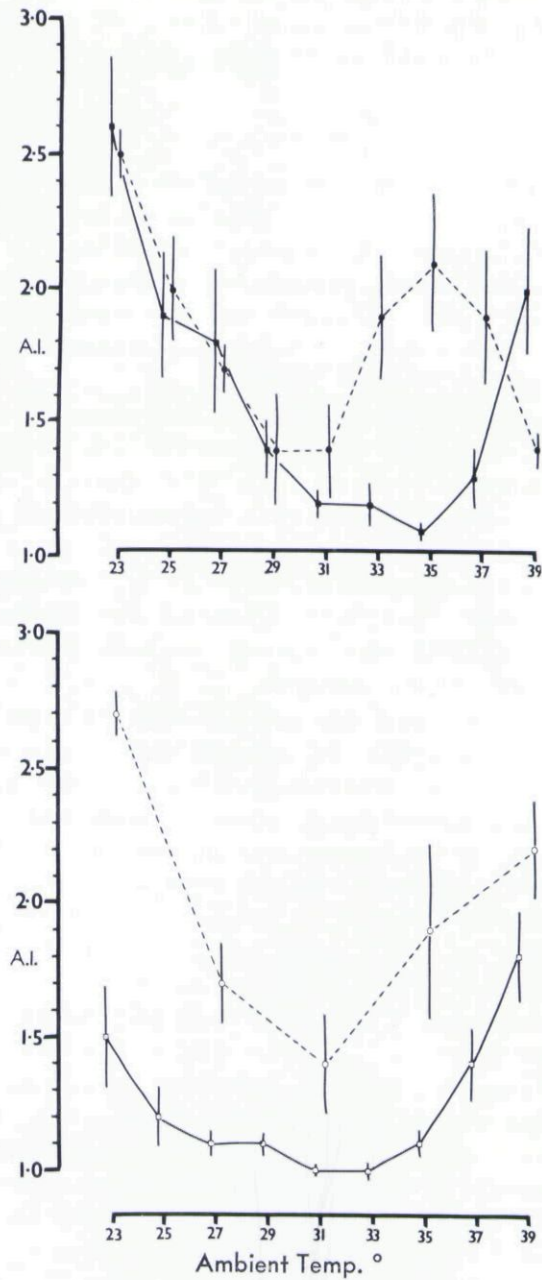


Fig. 3. Activity index A.I.: (mean $\pm$ S.E.M.) in response to different ambient temperatures in white and spiny mice.

● --- ● white mice fully-fed
 ■ --- ■ spiny mice fully-fed (upper)
 ○ --- ○ white mice fasted
 □ --- □ spiny mice fasted (lower)

Histology on the excised pancreas of 10 spiny and white mice sacrificed at the conclusion of all experiments showed the classical appearances of islets comprising 8-15% of pancreatic mass, compared with 1-2% in white mice (assessed by microscopic projection and tracing).

DISCUSSION.

The present findings parallel the original observations of Dawson (1955), who reported that basal oxygen consumption in the desert dwelling Merriam Kangaroo rat was 25% below that expected on the basis of Brody's (1945) formula. Using the deer mouse (*peromyscus*), Murie (1961) and McNab and Morrison (1963) have since suggested that this phenomenon has potential selective value for the avoidance of lethal body temperature and minimizing caloric requirements.

The mean minimum oxygen consumption of $0.64 \text{ ml g}^{-1} \text{ h}^{-1}$ for fully-fed spiny mice was 57% below that calculated on the basis of the Brody formula for the mean group weight ($1.44 \text{ ml g}^{-1} \text{ h}^{-1}$). Furthermore, the thermoneutral zone ($33^{\circ}\text{--}37^{\circ}$) of *Acomys* is higher than that recorded for any other desert rodent.

The mean oxygen consumption of $1.25 \text{ ml g}^{-1} \text{ h}^{-1}$ for white mice, although also below the expected value, is in agreement with data evaluated in this species by Morrison (1948) when adjusted for body weight, thus providing validation for the methodology employed.

The E-ratio distributions within groups reflect the slope of regression of activity on oxygen consumption. In order to obtain more meaningful comparisons, regression equations were calculated from individual paired oxygen and activity data. Having confirmed that the correlation coefficient (r) was significant ($p \leq 0.001$) for each group, linearity was not assessed, and data were simply expressed as the four single order regression equations below, and diagrammatically represented in Fig. 4:

(a) Fully-fed spiny mice $y = 0.48x + 0.3$ ($r = 0.76$)

(b) Fasted spiny mice $y = 0.26x + 0.32$ ($r = 0.52$)

(c) Fully-fed white mice $y = 0.56x + 0.59$ ($r = 0.67$)

(d) Fasted white mice $y = 0.48x + 0.47$ ($r = 0.82$)

The advantage of expressing the data in this way rests in the ability to utilize oxygen consumption readings at greater than basal activity. Thus, for groups a, c and d, the slopes are identical, only the y intercepts varying, and one can conclude that differences in E-ratio between these groups depend entirely on the differences in basal oxygen consumption, activity being associated with the expected proportional increments of oxygen consumption in each group. The regression line for fasting spiny mice, however, differs in having not only a lower y intercept, but also a flatter slope, shown by statistical comparison of regression coefficients to be highly significant ($t = 4.89$, $p = 0.001$). Thus, the spiny mouse, especially when fasted, appears to have a strikingly efficient energy conservation system.

There remained a possibility that the interspecies difference in oxygen consumption might be based on a smaller lean body mass, despite similar weights in spiny and white mice. However, whole body analysis for fat using the method of Folch, Lees and Sloane-Stanley (1957), in 4 spiny and 4 white mice was carried out on completion of the metabolic experiments. Mean body weight (and percentage body fat) was 49.1 g (23.2%) and 47.4 g (23.0%) in white and spiny mice respectively, similar to the spiny mouse data established by Gonet *et al.* (1966).

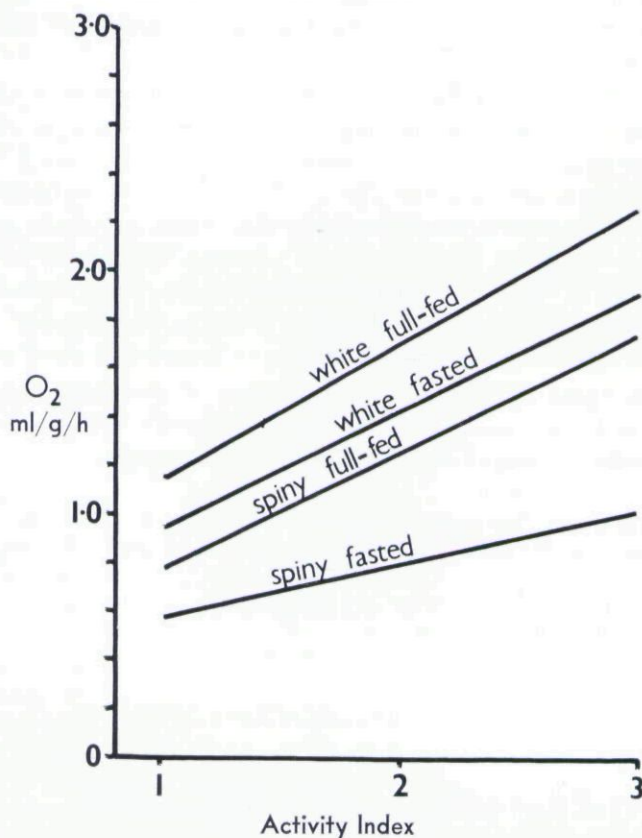


Fig. 4. Calculated regression lines relating paired data for oxygen consumption and activity index.

The association of low metabolic rate with a tendency to diabetes is almost certainly not coincidental. In the obese-hyperglycaemic (ob/ob) mutant mouse, where a peripheral resistance to insulin action has been demonstrated in the homozygous state, Mayer *et al.* (1952) have recorded unexpectedly low minimum oxygen consumption rates and significant mortality when these animals are cold exposed, based on an inability to increase heat production. That this was genetic in origin was shown by Fried and Antopol (1963), who demonstrated that, when assessed in early post-natal life, when no weight trends were yet

apparent, those animals destined to become obese and hyperglycaemic already had significantly lower metabolic rates. Berti (1963), in rats rendered artificially diabetic with alloxan, demonstrated a decrease in oxygen consumption and rectal temperature, together with a failure to increase oxygen consumption in response to cold.

Finally, in man, certain aboriginal tribes in South-Central Australia have been shown to have unexpectedly low basal oxygen consumption and an inability to increase heat production in response to cold (Hammel *et al.*, 1969). This same group of tribes has recently been shown to have a high prevalence of diabetes exceeding a similarly tested group of Australian Europeans by a factor of ten (Wise *et al.*, 1976).

It has been suggested by Neel (1962) that persistence of the apparently lethal diabetic genotype in human populations is based on a peculiarly "thrifty" gene or group of genes, which may have had survival advantage under certain environmental conditions, and this concept has been analysed in detail elsewhere (Wise, 1968). The present studies suggest that a low metabolic rate represents the common denominator for this survival advantage. Thus, one or more genetic defects resulting in "defective" glucose utilization may have been highly significant in the evolution of species inhabiting warm and arid areas, in which the overt diabetic syndrome was prevented by environmental food scarcity.

Further support for the survival advantage of low metabolic rates in the spiny mouse will be subsequently presented, together with an analysis of the possible evolutionary significance of the diabetic genotype.

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